

Tzu-Heng (Brian) Huang

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🌐 [zihengh1.github.io](https://github.com/zihengh1)

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Education

- 2021 – 2026 📖 **Ph.D. in Computer Science. University of Wisconsin-Madison.**
Advised by Frederic Sala.
- 2016 – 2020 📖 **B.S. in Computer Science. National Chengchi University.**
Advised by Man-Kwan Shan and Ling-Jyh Chen. Major GPA: 3.96/4.00.

Research Summary

My research focuses on *data-centric AI for multimodal models*, with the goal of enabling ML systems to learn more from less through cheaper, higher-quality supervision. Several *data lifecycle* projects I have worked on: (i) *rubric-guided RL* for high-quality captioning sources used in visual pretraining; (ii) *data selection* via weight-space geometry for efficient pretraining (**ICML'26 + ICML'25 DataWorld Oral**); (iii) *online data mixing* to reduce overhead in multimodal training; (iv) *data curation* via ensemble methods and objective detection (**1st place on the DataComp'23 leaderboard**); (v) a $500\times$ cheaper *automated labeling* system as an alternative to LLM annotators (**NeurIPS'24 Spotlight**); and (vi) a *hybrid evaluation framework* with program synthesis that pushes beyond the current LLM-judge Pareto frontier.

Research Experience

- Oct. 2025 – Jan. 2026 📖 **AIML Research Intern. Apple Inc.**
advised by Manjot Bilkhu and Javier Movellan.
— *Rubric-based RL for Dense Captioning Tasks.*
- May. 2025 – Sep. 2025 📖 **Research Scientist Intern. Meta GenAI (now MSL).**
advised by Ernie Chang, Sang Michael Xie, Yiting Lu, and David Kant.
— *Learnability-aware Synthetic Data Generation.*
- May. 2024 – Dec. 2024 📖 **AIML Research Intern. Apple Inc.**
advised by Javier Movellan and Manjot Bilkhu.
— *Data Selection via Weight-space Geometry for Efficient Pretraining.*
— *Optimizing Domain Mixtures for MLLM Pretraining.*
- Aug. 2021 – Aug. 2026 📖 **Graduate Research Student. UW-Madison.**
advised by Frederic Sala.
— *Data-centric AI for Foundation Models: Auto-labeling and Data Curation.*
— *Parameter Marketplace: Through Model Merging and Auction Agents.*
- May. 2023 – May. 2024 📖 **Founder. Awan.AI LLC.**
— *Tongue Syndrome Diagnosis and LLM for Traditional Chinese Medicine.*
— *Automating TCM Diagnosis: Herbal-based Recommendation System.*
- Jun. 2019 – Sep. 2019 📖 **Research Intern. Argonne National Laboratory.**
advised by Charlie Catlett and Rajesh Sankaran.
— *Ensemble-based Time Series Calibration for Low-cost Sensors (Array of Things).*
- Sep. 2018 – Aug. 2021 📖 **Research Assistant. National Chengchi University.**
advised by Man-Kwan Shan.
— *Spatio-temporal Modeling in Large-scale Sensor Networks.*
- Feb. 2018 – Jul. 2020 📖 **Research Intern. Academia Sinica.**
advised by Ling-Jyh Chen.
— *Large-scale Air Quality Sensor Networks (AirBox).*

Publications and Preprints

- 1 **T.-H. Huang**, S. Qiu, and F. Sala, “Codifying the Judge: Scalable Evaluation via Program Distillation,” Under Review, 2026.
- 2 **T.-H. Huang**, A. Goyal, J. Cooper, and F. Sala, “WARP: Weight-Space Analysis for Recovering Training Data Portfolios,” in *ICML Workshop: Weight-Space Symmetries: from Foundations to Practical Applications (WSS)*, 2026. 🔗 URL: <https://openreview.net/pdf?id=5GJDmHFNUY>
- 3 N. Roberts, S. Cho, Z. Gao, **T.-H. Huang**, A. Wu, G. Orlanski, A. Trost, K. Buchanan, A. Albarghouthi, and F. Sala, “Test-Time Scaling Makes Overtraining Compute-Optimal,” Under Review, 2026. 🔗 URL: <https://arxiv.org/pdf/2604.01411>
- 4 **T.-H. Huang**, S. Salekin, J. Movellan, F. Sala, and M. Bilkhu, “RubiCap: Rubric-Guided Reinforcement Learning for Dense Image Captioning,” Under Review, 2026. 🔗 URL: <https://arxiv.org/abs/2603.09160>
- 5 J. Zhao, C. Shin, **T.-H. Huang**, S. S. S. Namburi, and F. Sala, “CARE: Confounder-Aware Aggregation for Reliable LLM Evaluation,” in *International Conference on Machine Learning (ICML)*, 2026. 🔗 URL: <https://arxiv.org/abs/2603.00039>
- 6 J. Saad-Falcon, E. K. Buchanan, M. F. Chen, **T.-H. Huang**, B. McLaughlin, T. Bhathal, S. Zhu, B. Athiwaratkun, F. Sala, S. Linderman, A. Mirhoseini, and C. Re, “Shrinking the Generation-Verification Gap by Scaling Compute for Verification,” in *Neural Information Processing Systems (NeurIPS)*, *ICML Workshop: Efficient Systems for Foundation Models (ES-FoMo III)*, and *ICML Workshop: Multi-Agent Systems in the Era of Foundation Models: Opportunities, Challenges and Futures (MAS)*, 2025. 🔗 URL: <https://arxiv.org/abs/2506.18203>
- 7 **T.-H. Huang**, H. Vishwakarma, and F. Sala, “Time to Impeach LLM-as-a-Judge: Programs are the Future of Evaluation,” in *ICML Workshop: Programmatic Representations for Agent Learning (PRAL)*, 2025. 🔗 URL: <https://arxiv.org/abs/2506.10403>
- 8 J. Zhao, C. Shin, **T.-H. Huang**, S. S. S. Namburi, and F. Sala, “From Many Voices to One: A Statistically Principled Aggregation of LLM Judges,” in *NeurIPS Workshop: Evaluating the Evolving LLM Lifecycle: Benchmarks, Emergent Abilities, and Scaling*, and *NeurIPS Workshop: Reliable ML from Unreliable Data*, 2025. 🔗 URL: <https://openreview.net/pdf?id=0u53DNvjx7>
- 9 A. Ge, **T.-H. Huang**, J. Cooper, A. Trost, Z. Chu, S. S. S. Namburi, Z. Cai, K. Park, N. Roberts, and F. Sala, “R&B: Domain Regrouping and Data Mixture Balancing for Efficient Foundation Model Training,” in *ICML Workshop: Unifying Data Curation Frameworks Across Domains (DataWorld)*, and *ICML Workshop: Data in Generative Models (The Bad, the Ugly, and the Greats) (DIG-BUGS)*, 2025. 🔗 URL: <https://arxiv.org/abs/2505.00358>
- 10 **T.-H. Huang**, M. Bilkhu, J. Cooper, F. Sala, and J. Movellan, “Evaluating Sample Utility for Efficient Data Selection by Mimicking Model Weights,” in *International Conference on Machine Learning (ICML)*, and *ICML Workshop: Unifying Data Curation Frameworks Across Domains (DataWorld)* [**Oral Paper**], 2025. 🔗 URL: <https://arxiv.org/abs/2501.06708>
- 11 **T.-H. Huang**, C. Cao, V. Bhargava, and F. Sala, “The ALCHEmist: Automated Labeling 500x CHEaper than LLM Data Annotators,” in *Neural Information Processing Systems (NeurIPS)* [**Spotlight Paper (Top 2.08%)**], 2024. 🔗 URL: <https://arxiv.org/abs/2407.11004>
- 12 W. Tan, N. Roberts, **T.-H. Huang**, J. Zhao, J. Cooper, S. Guo, C. Duan, and F. Sala, “MoRe Fine-Tuning with 10x Fewer Parameters,” in *ICML Workshop: Efficient Systems for Foundation Models (ES-FoMo)*, and *ICML Workshop: Foundation Models in the Wild.*, 2024. 🔗 URL: <https://arxiv.org/abs/2408.17383>
- 13 N. Roberts, X. Li, D. Adila, S. Crompt, **T.-H. Huang**, J. Zhao, and F. Sala, “Geometry-Aware Adaptation for Pretrained Models,” in *Neural Information Processing Systems (NeurIPS)*, 2023. 🔗 URL: <https://arxiv.org/abs/2307.12226>

- 14 **T.-H. Huang**, H. Vishwakarma, and F. Sala, “Train ’n Trade: Foundations of Parameter Markets,” in *Neural Information Processing Systems (NeurIPS)*, 2023.  URL: <https://arxiv.org/abs/2312.04740>
- 15 **T.-H. Huang**, C. Shin, S. J. Tay, D. Adila, and F. Sala, “Multimodal Data Curation via Object Detection and Filter Ensembles,” in *ICCV Workshop: Towards the Next Generation of Computer Vision Datasets (TNGCV) [1st place on the Datacomp leaderboard (small-scale filtering track)]*, 2023.  URL: <https://arxiv.org/abs/2401.12225>
- 16 **T.-H. Huang**, C. Cao, S. Schoenberg, H. Vishwakarma, N. Roberts, and F. Sala, “ScriptoriumWS: A Code Generation Assistant for Weak Supervision,” in *ICLR Workshop: Deep Learning For Code (DL4C)*, 2023.  URL: <https://arxiv.org/abs/2502.12366>
- 17 N. Roberts, X. Li, **T.-H. Huang**, D. Adila, S. Schoenberg, C.-Y. Liu, L. Pick, H. Ma, A. Albarghouthi, and F. Sala, “AutoWS-Bench-101: Benchmarking Automated Weak Supervision with 100 Labels,” in *Neural Information Processing Systems (NeurIPS)*, 2022.  URL: <https://arxiv.org/abs/2208.14362>
- 18 **T.-H. Huang**, C.-H. Tsai, and M.-K. Shan, “Key Sensor Discovery for Quality Audit of Air Sensor Networks,” in *ACM International Conference on Mobile Systems, Applications, and Services (MobiSys)*, 2020.  URL: <https://dl.acm.org/doi/abs/10.1145/3386901.3396606>

Miscellaneous

Awards

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| 2026 |  Gold Reviewer Award , recognized by ICML’26. |
| 2025 |  Oral Paper: Grad-Mimic (Over 100+ submissions) , selected by ICML’25 DataWorld. |
| 2024 |  Spotlight Paper (Top 2.08%): The Alchemist , selected by NeurIPS’24. |
| 2023 |  ICCV Datacomp Competition , won the first place in the small-scale filtering track. |
| |  Scholar Award , granted by NeurIPS’23. |
| 2021 |  First-year Departmental Scholarship , granted by UW-Madison. |
| 2020 |  Research Intern Scholarship , granted by National Chengchi University. |
| |  Undergrad Research Scholarship , granted by Ministry of Science and Technology. |

Invited Talks

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| Jan. 2026 |  Data Recipes: Labeling, Selection, and Verification , invited by National Tsing Hua University (NTHU). |
| |  Data Recipes: Labeling, Selection, and Verification , invited by National Taiwan University (NTU). |
| |  Data Recipes: Labeling, Selection, and Verification , invited by Taiwan Semiconductor Manufacturing Company (TSMC). |
| |  Data Recipes: Labeling, Selection, and Verification , invited by National Yang Ming Chiao Tung University (NYCU). |
| Aug. 2025 |  Data Recipes: Automated Labeling and Efficient Selection , invited by Scaling Intelligence Lab (Azalia Mirhoseini’s group) in Stanford University. |
| Apr. 2025 |  Spatio-temporal Modeling for Underwater Sensor Networks , invited by National Taipei University of Technology (NTUT). |
| Dec. 2019 |  Air Quality Sensor Network Developments , invited by Nangang High School (Taiwan). |
| Sep. 2019 |  Internship Research Talk , invited by National Chengchi University. |
| Jul. 2019 |  LASS Conference: International Session , invited by Academia Sinica. |
| Mar. 2019 |  Techbang Magazine: PiM25 Project , invited by Techbang Magazine. |

Miscellaneous (continued)

- Jan. 2019  **Raspberry Pi Jam: PiM25 Project**, invited by Raspberry Pi Foundation (Taiwan).
- Jan. 2019  **Raspberry Pi Meetup: PiM25 Project**, invited by Raspberry Pi Foundation (Taiwan).

Academic Services

- 2026 – Present  **Organizer**, Data Foundations of AI (**a new data-first online seminars**).
- 2021 – Present  **Active Paper Reviewer**, TMLR, NeurIPS, ICLR, and ICML.
- 2023  **Co-organizer**, AutoML Cup in AutoML Conference.
- 2022 – 2023  **President of Student Association of Taiwan**, UW-Madison.
- 2021 – 2022  **Vice President of Student Association of Taiwan**, UW-Madison.

Consulting Roles

- 2025  **Research Consultant**, Water Resources Agency, Ministry of Economic Affairs in Taiwan. Focusing on *underwater sensor network developments* and *flood detection modeling*.
- 2023 – 2025  **Research Consultant**, TechTCM. Focusing on *AI for Traditional Chinese Medicine*.